Cole Becker

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MIT PhD candidate working in measure transport, diffusion models, and optimization. I am primarily interested in principled methods for accelerating and adapting generative models without retraining: models that are faster (fewer oracle evaluations), more flexible (adaptable to reward-specific objectives), and data-aware (robust to varying data sources and quality). My current work designs diffusion samplers by jointly choosing the noise schedule and solver via optimal rational approximation in the score oracle, mirroring how Chebyshev acceleration optimizes over polynomials in the gradient oracle. I am seeking an applied research role in generative AI for 2027.

EDUCATION

Massachusetts Institute of Technology Sep. 2023 – Present
PhD Candidate in Computer Science – Advisor: Pablo Parrilo Cambridge, MA

Princeton University Sep. 2017 – May 2023
M.Eng. and B.S.E. in Electrical and Computer Engineering Princeton, NJ
Advisors: Bartolomeo Stellato and Amir Ali Ahmadi

- **GPA**: 3.95/4.0 - *Magna Cum Laude*
- **Thesis**: Learning Uncertainty Sets in Robust Optimization

RESEARCH

Elliptic Diffusion Sampling Summer 2026 – Present
Massachusetts Institute of Technology – Professor Pablo Parrilo Cambridge, MA

- Designed a training-free diffusion sampler that pairs a new *elliptic* noise schedule with a matched discretization, chosen jointly using optimal (Zolotarev) rational approximation.
- Proved the sampler is minimax optimal for Gaussian targets among fixed-schedule score-based solvers, and showed that the schedule improves standard solvers at low step counts, from synthetic benchmarks to ImageNet image generation.

Steering Pretrained Diffusion Models Fall 2025 – Present
Massachusetts Institute of Technology Cambridge, MA
with Professor Pablo Parrilo and Professor Asu Özdağlar

- Exploring training-free methods for steering pretrained diffusion and flow models toward rewards and inverse-problem constraints (reward tilting, posterior sampling), comparing guidance-based samplers such as DPS with interpolant-based alternatives.

Sim-to-Real Transfer for Diffusion Policies Fall 2025 – Present
Massachusetts Institute of Technology - Professor Russ Tedrake Cambridge, MA

- Studying how to train robot-manipulation diffusion policies from abundant simulated and scarce real demonstrations, using flow models to transport simulated data toward the real-data distribution.

Robust Optimization 2022 - 2023
Princeton University - Advised by Professor Bartolomeo Stellato Princeton, NJ

- Co-led a project to build a robust extension package to cvxpy to automate dualization from robust optimization problems to tractable convex programs.

PUBLICATIONS

Mean Robust Optimization 2023
I. Wang, C. Becker, B. Van Parys, B. Stellato

- *Mathematical Programming* 213 (1), 1235-1277
- **Best Paper Award** ICCOPT 2022

Learning Decision-Focused Uncertainty Sets in Robust Optimization 2023
I. Wang, C. Becker, B. Van Parys, B. Stellato

- arXiv preprint arXiv:2305.19225

TEACHING

Instructor – Diffusion Models Jan. 2025, 2026
Massachusetts Institute of Technology Cambridge, MA

- Co-taught introductory January course in Diffusion Models, lectured, designed problem sets, and led office hours.

Assistant Instructor – Convex Optimization and Other Courses Fall 2020 – Spring 2022
Princeton University Princeton, NJ

- Served as Assistant Instructor and Graduate/Undergraduate TA for multiple courses including Convex Optimization, Convex & Conic Optimization (PhD), and Introduction to Logic Design. Responsibilities included leading weekly office/study hours, grading and writing assignments and exams, delivering review lectures

FELLOWSHIPS AND AWARDS

National Science Foundation Graduate Research Fellowship (NSF GRFP) 2023

George Mueller Award – Princeton University 2022
Highest achievement across engineering and athletics

TALKS

ICCOPT July 2022
Learning Uncertainty Sets in Robust Optimization Lehigh University

SKILLS

Software : Python, MATLAB, CVXPY, PyTorch, JAX, NumPy, Pandas, Seaborn, Matplotlib, Java
Languages : English (native), French (B1), Spanish (B1)

ACTIVITIES

Squash 2008 – Present

- Princeton Varsity Squash Team – Captain (2020–21)
- Professional Squash Player (Top 5 U.S.)
- U.S. National Squash Team Member (2015–2021)